

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460957.7517 +/- 0.0071 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.062
Transit depth (Rp/Rs)^2	3.93 +/- 2.45 [%]
Orbital Inclination (inc)	88.41 +/- 1.9
Airmass coefficient 1 (a1)	1.74 +/- 1.09
Airmass coefficient 2 (a2)	-0.5 +/- 1.44
Scatter in the residuals of the lightcurve fit is	65.03 %
Best Comparison Star	#2 - [58, 421]
Optimal Aperture	1.58
Optimal Annulus	4.0
Transit Duration (day)	0.089 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.198 ± 0.062 vs 0.1592 (deviation 0.47σ)
Duration fit vs published	0.089 d vs 0.0946 d (deviation 0.28σ)
Mid-time O–C	17.7 min
Depth SNR	2.4
Residual scatter	65.03 %

Light curve

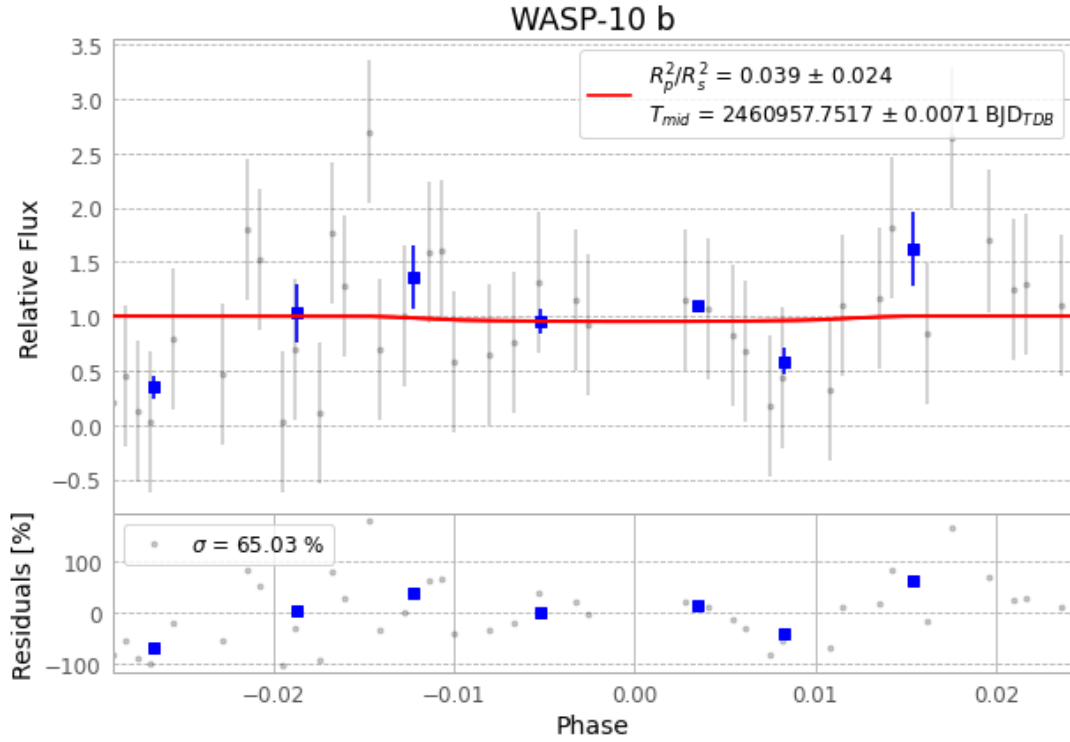


Figure 1 — FinalLightCurve_WASP-10 b_2025-10-08.png (SHA-256 e8868b8c5d3fb5c4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [391, 117], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bc33e55ecffbe89b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 574e53e

Data provenance

81 FITS frames (53 MB), WASP-10251009034515.FITS → WASP-10251009074515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-251009.log (5e49c96245b5...), FinalLightCurve_WASP-10 b_2025-10-08.png (e8868b8c5d3f...), FinalLightCurve_WASP-10 b_2025-10-08.csv (3d3e76220a2b...)

Compute resources

Wall time 180.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 18:01Z.